

MI-29: Resolution

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11 September 2026

Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-29: an exact counterexample to the arbitrary-exponent determinant extension

The repository's extension asks whether

$$\det(A^k + |AB|^p) \geq \det(A^k + |BA|^p) \quad (1)$$

holds for $A > 0$, invertible Hermitian B , and $k, p \geq 0$. The established $k = 2$ inequality in [1] is a different, more restricted assertion; the example below does not contradict it.

Theorem 1.1. *Inequality (1) is false for real matrices of order three, with $k = 6$ and $p = 8$. One may take*

$$A = \text{diag}\left(2, 1, \frac{1}{2}\right), \quad B = \frac{1}{5} \begin{pmatrix} -1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix}.$$

Proof. The matrix A is positive definite, and B is real symmetric with $\det B = -1/125$, so all hypotheses are met. Write $M = 5B$ and

$$D = A^6, \quad H = (MA^2M)^4, \quad J = (AM^2A)^4.$$

Since $|AB|^2 = BA^2B$ and $|BA|^2 = AB^2A$, the left- and right-hand sides of (1) are respectively

$$\det(D + 5^{-8}H), \quad \det(D + 5^{-8}J).$$

In dimension three, expansion by multilinearity gives

$$\det(D + zK) = \det D + z \operatorname{tr}(\operatorname{adj}(D)K) + z^2 \operatorname{tr}(\operatorname{adj}(K)D) + z^3 \det K.$$

Here $\det H = \det J$. Direct rational matrix multiplication therefore yields the following two-coefficient certificate:

$$\det(D + zJ) - \det(D + zH) = \frac{2089017}{16}z - \frac{31188746592549}{1024}z^2. \quad (2)$$

Substituting $z = 5^{-8}$ into (2) gives

$$\det(A^6 + |BA|^8) - \det(A^6 + |AB|^8) = \frac{21036678407451}{156250000000000} > 0,$$

which is the reverse of the proposed inequality. In particular, the individual exact determinants are

$$\det(A^6 + |AB|^8) = \frac{136990346414301954149}{61035156250000000000},$$

$$\det(A^6 + |BA|^8) = \frac{4537743716162890657}{1907348632812500000}.$$

The verifier checks both the polynomial certificate and these values independently. $\square$

Remark 1.2 (Hypothesis distinction). The counterexample uses an indefinite B , which is permitted by the stated Hermitian-input extension. It does not settle a version imposing $B > 0$, nor does it alter the known $k = 2$ theorem. The precise target statement should be retained in any repository status update.

References

- [1] Mohammad M. Ghabries, Hassane Abbas, and Bassam Mourad. On some open questions concerning determinantal inequalities. *Linear Algebra and its Applications*, 596:169–183, 2020. The established $k = 2$ case; not the arbitrary- k repository extension. [doi:10.1016/j.laa.2020.03.009](https://doi.org/10.1016/j.laa.2020.03.009).